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$$\begin{aligned}f(x) &= \log_x 2 = \frac{\ln 2}{\ln x} \\ \frac{d}{dx} \left(\frac{\ln 2}{\ln x} \right) &= \ln 2 \cdot \frac{d}{dx} \left(\frac{1}{\ln x} \right) \\ &= \ln 2 \cdot \frac{d}{dx} ((\ln x)^{-1}) \\ &= \ln 2 \cdot \frac{-1}{(\ln x)^2} \cdot \frac{1}{x} \\ &= \frac{-\ln 2}{x(\ln x)^2}\end{aligned}$$

References